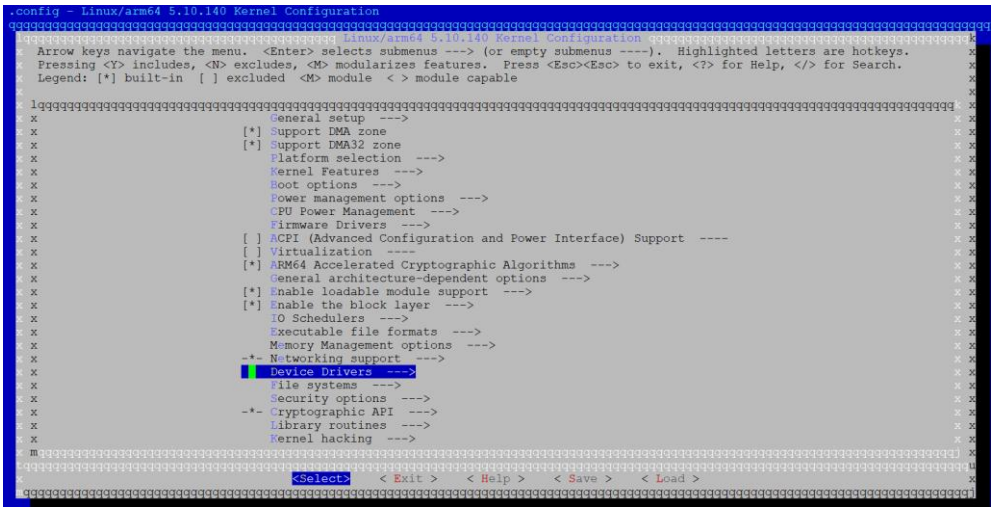
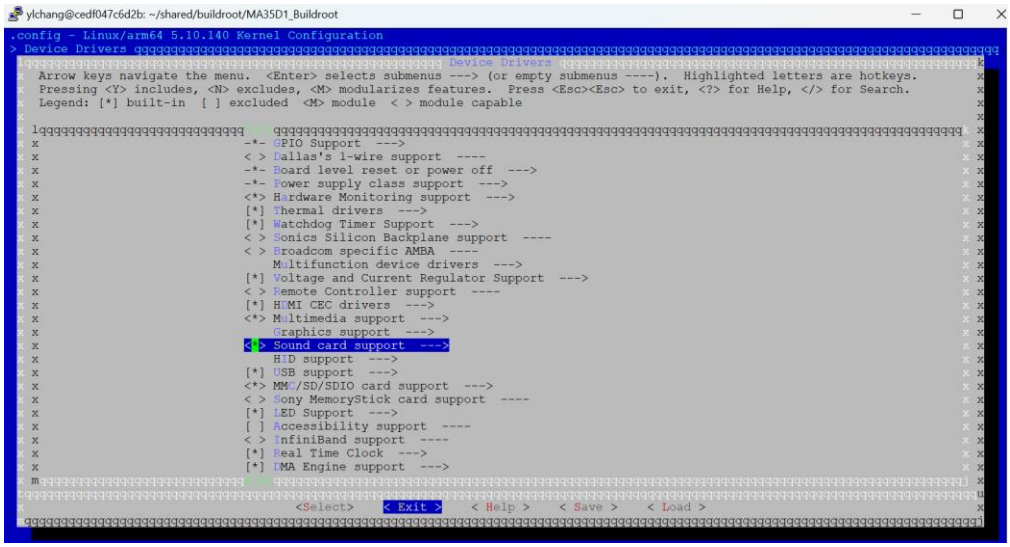
Simple-Card Bring-up Guide

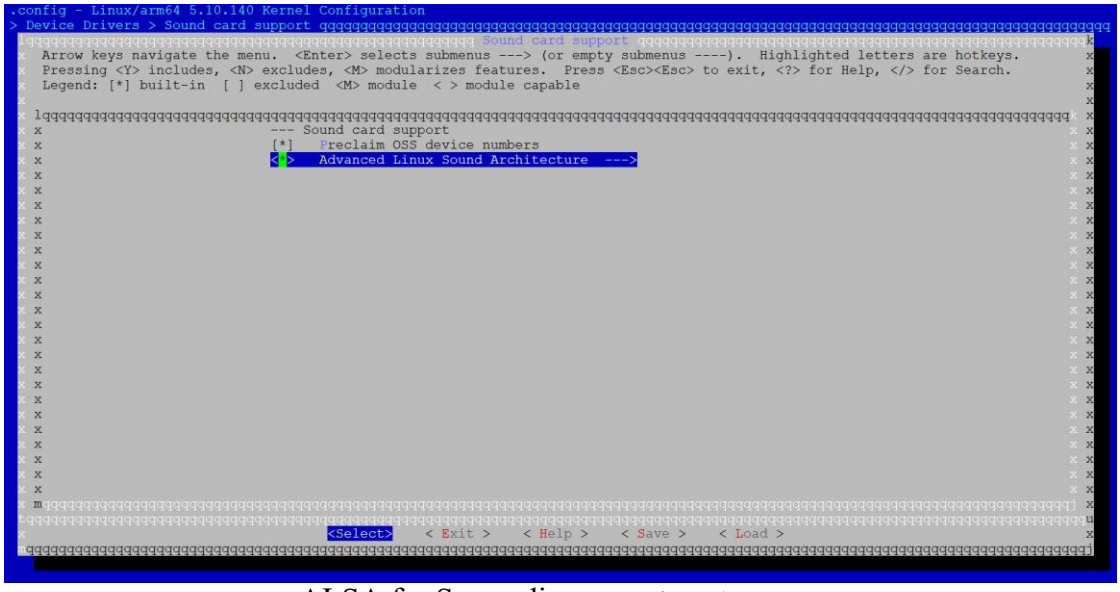
- **Introduction**
  - The **ALSA SoC (ASoC) Simple-Card** is a lightweight and generic machine driver used to bind CPU and codec components through Device Tree configuration. In the ASoC architecture, there are three essential components: the **Machine Driver**, **Platform Driver**, and **Codec Driver**. The simple-card is useful for quick bring-up. It serves as a generic **Machine Driver** replacement. Achieve function development quickly with minimal effort.
- **Linux Simple-Card Configuration**
  - First, you should enable Simple-Card Support
    - To use `simple-card` in your Linux system, you must first ensure the driver is enabled in the kernel configuration (`menuconfig`).
    - make linux-menuconfig
      - Device Drivers - - ->



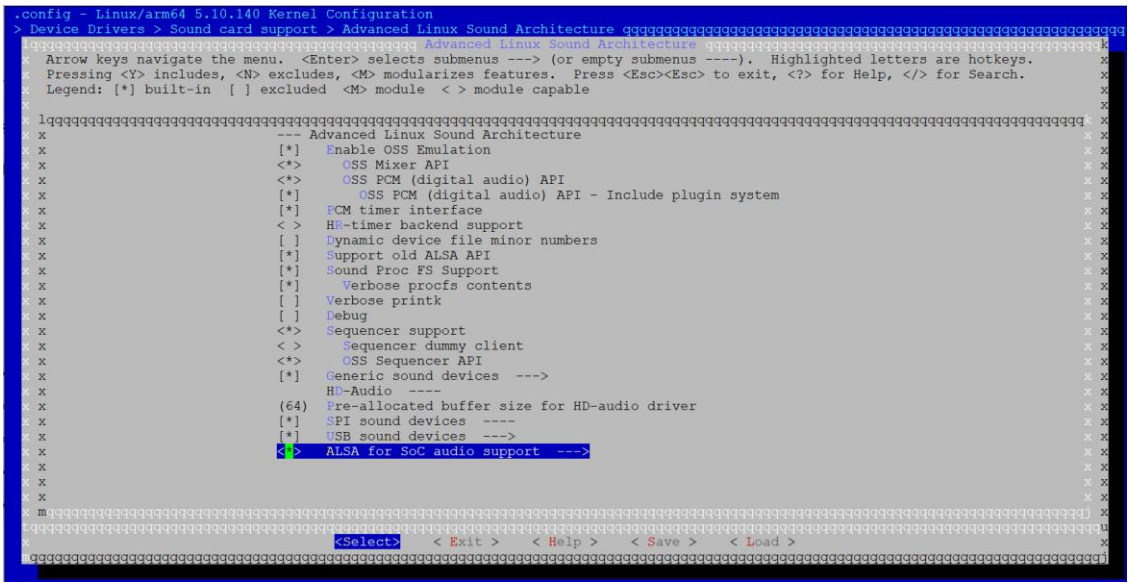
      - <*> Sound card Support - - ->



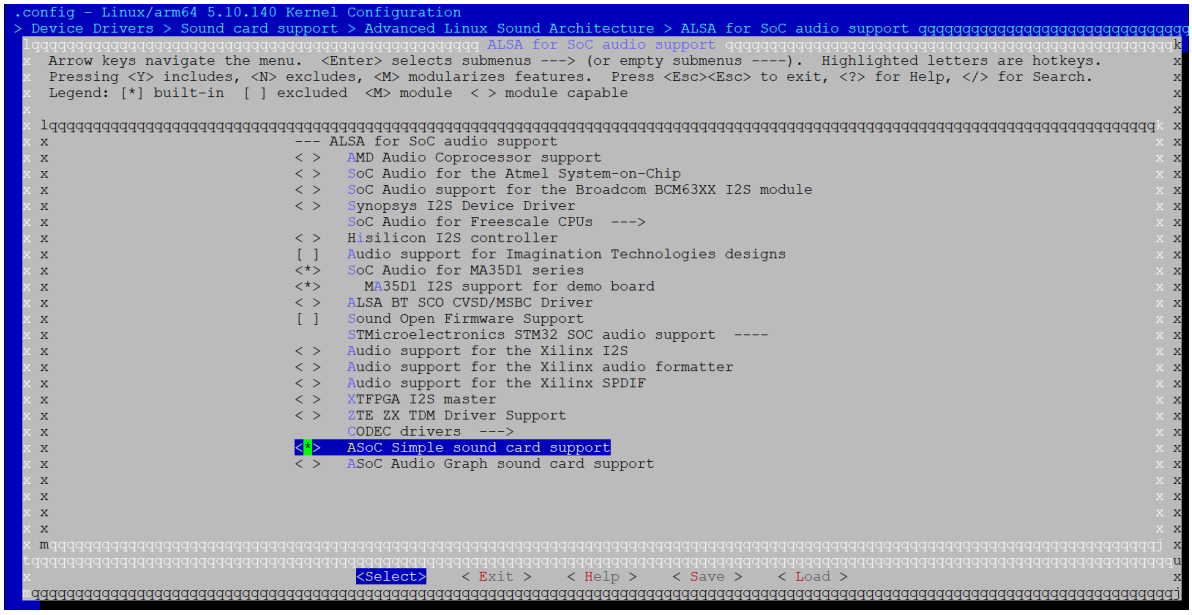
- Advanced Linux Sound Architecture - - ->



- ALSA for Soc audio support - - ->



- <*> ASoc Simple sound card support
  - Press “Space” and Select ASoc Simple sound card support
  - Finally, exit and save configuration, then build the kernel image.



- **Device Tree Configuration for Simple-Card**

  To use Simple-Card, the following components must be defined in the Device Tree:

- **Sound Node:** The top-level node that binds CPU and codec via DAI links.
  - Required properties:
    - **- compatible:** “simple-audio-card.
  - Optional properties:
    - **- simple-audio-card,name:** User specified audio sound card name.
    - **- simple-audio-card,widgets:** Machine DAPM widgets
    - **- simple-audio-card,routing:** A list of the connections between audio components. Each entry is a pair of strings, the first being the connection's sink, the second being the connection's source.
    - **- simple-audio-card,mclk-fs:** Multiplication factor between stream rate and codec mclk. When defined, mclk-fs property defined in dai-link sub nodes are ignored.
    - **- simple-audio-card,format:** CPU/CODEC common audio format. "i2s", "right_j", "left_j" , "dsp_a"
      "dsp_b", "ac97", "pdm", "msb", "lsb"
    - **- simple-audio-card,bitclock-master:** Indicates dai-link bit clock master. phandle to a cpu or codec subnode.
    - **- simple-audio-card,frames-master:** Indicates dai-link frame master. phandle to a cpu or codec subnode.
    - **- system-clock-direction-out:** pecifies clock direction as 'out' on initialization. It is useful for some aCPUs with fixed clocks.
    - **- simple-audio-card,hp-det-gpio:** Reference to GPIO that signals when headphones are attached.
    - **- simple-audio-card,mic-det-gpio:** Reference to GPIO that signals when a microphone is attached.
    - **- simple-audio-card,aux-devs:** List of phandles pointing to auxiliary devices, such as amplifiers, to be added to the sound card.
  - Required dai-link subnodes:
    - **CPU Node:**
      - **- simple-audio-card,cpu:** Usually refers to the I2S controller node.
    - **Codec Node:**
      - **- simple-audio-card,codec:** Refers to the audio codec, such as NAU8822.
- For more detailed attribute descriptions, refer to:
  - https://www.kernel.org/doc/Documentation/devicetree/bindings/sound/simple-card.txt

- **Example: Device Tree Configuration for NAU8822 Codec**

```
//ma35d1.dtsi
clk: clock-controller@40460200 {
        compatible = "nuvoton,ma35d1-clk", "syscon", "simple-mfd";
        reg = <0x00000000 0x40460200 0x0 0x100>;
        #clock-cells = <1>;
        assigned-clocks = <&clk CAPLL>,
                          <&clk DDRPLL>, <&clk APLL>,
                          <&clk EPLL>, <&clk VPLL>;
        assigned-clock-rates =<1000000000>,
                          <266000000>, <196608000>, /* APLL Clock is set to
196608000 Hz */
                          <500000000>, <102000000>;
        clock-pll-mode = <0>, <1>, <0>, <0>, <0>;
        nuvoton,sys = <&sys>;
};

i2s0: i2s@40480000 {
        compatible = "nuvoton,ma35d1-audio-i2s";
        reg = <0x0 0x40480000 0x0 0x10000>;
        interrupts = <GIC_SPI 70 IRQ_TYPE_LEVEL_HIGH>;
        clocks = <&clk I2S0_GATE>;
        resets = <&reset MA35D1_RESET_I2S0>;
        pdma_reqsel_tx = <86>;
        pdma_reqsel_rx = <87>;
        mclk_out = <12288000>; /* mclk_out is set to 12288000 Hz */
        status = "disable";
};
//ma35d1-iot-512m.dts
&i2c1 {
        status = "okay";
        pinctrl-names = "default";
        pinctrl-0 = <&pinctrl_i2c1>;

        #address-cells = <1>;
        #size-cells = <0>;
        nau8822: nau8822@1a {
                compatible = "nuvoton,nau8822";
                #sound-dai-cells = <0>; /* It's Essential */
                reg = <0x1a> ;
                nuvoton,spk-btl;
                status = "okay";
        };
};

&i2s0 {
        status = "okay";
        pinctrl-names = "default";
        pinctrl-0 = <&pinctrl_i2s0>;
        #sound-dai-cells = <0>; /* It's Essential */
        powerdown-gpios = <&gpioc 2 GPIO_ACTIVE_HIGH>; /* PC2 */
};
```

```
&i2s_pcm {
        status = "disabled";
};

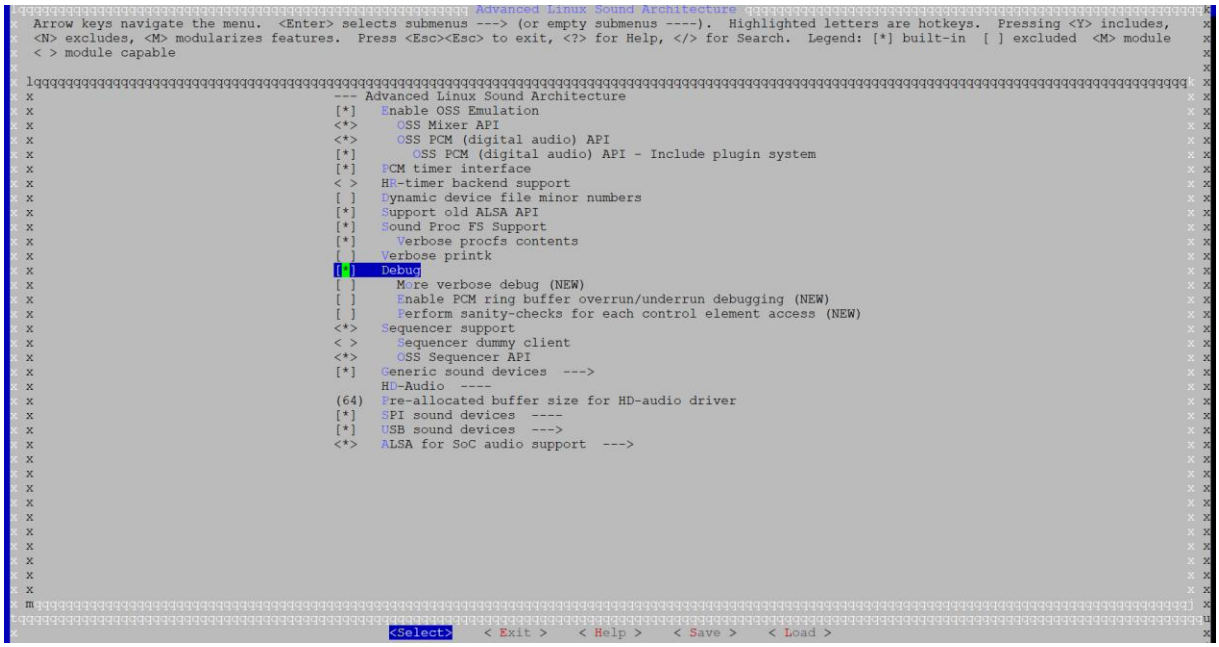
&sound {
        compatible = "simple-audio-card";
        simple-audio-card,name = "nau8822-hifi";
        simple-audio-card,format = "i2s";
        //simple-audio-card,bitclock-master = <&i2s0>;
        //simple-audio-card,frame-master = <&i2s0>;
        simple-audio-card,mclk-fs = <256>;

        status = "okay";
        simple-audio-card,widgets = "Headphone", "Headphone Jack",
                                    "Microphone", "Microphone Jack";
        simple-audio-card,routing = "Headphone Jack", "LHP",
                                    "Headphone Jack", "RHP",
                                    "LMICN", "Microphone Jack",
                                    "LMICP", "Microphone Jack";
        simple-audio-card,cpu {
                sound-dai = <&i2s0>;
        };
        simple-audio-card,codec {
                sound-dai = <&nau8822>;
        };
};
```

Note: 1. If you enable **MA35D1**'s "Simple-Card", **i2s_pcm** should be disabled.

2. The **#sound-dai-cells** property is essential in DAI nodes such as `i2s0` and `nau8822`

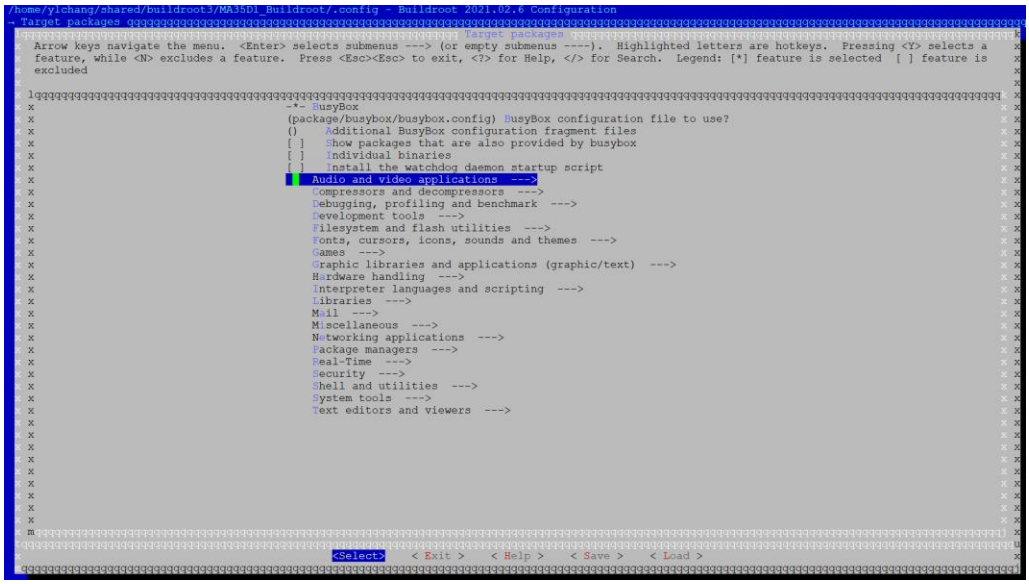
- **How to check if Simple-Card is working**
  - Enable Debug Mode



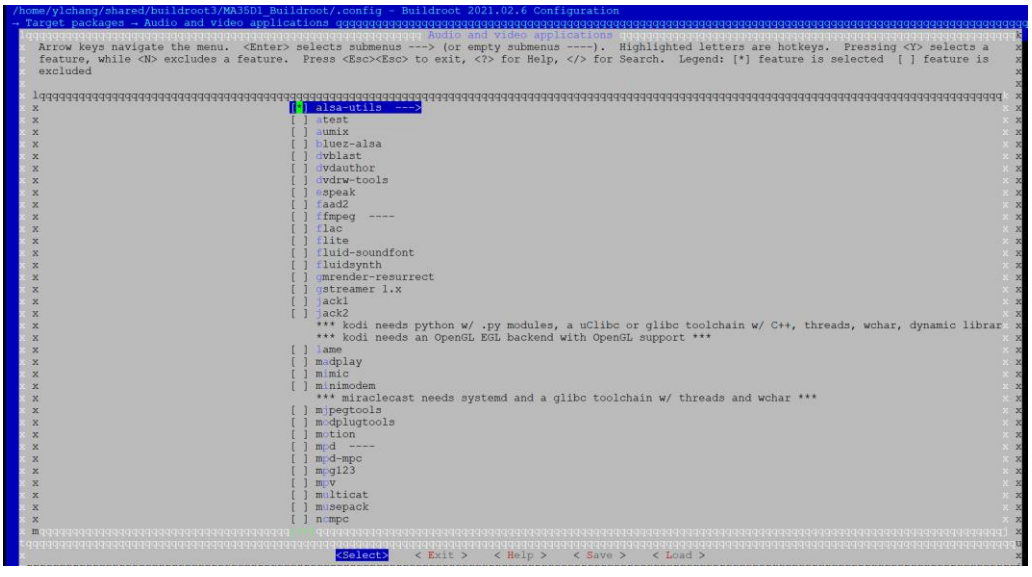
- dmesg (capture kernel log)
  - Checking if the ALSA device list is correct when the device is successfully registered.

```
[    3.355569] ALSA device list:
[    3.358581]   #0: nau8822-hifi
```

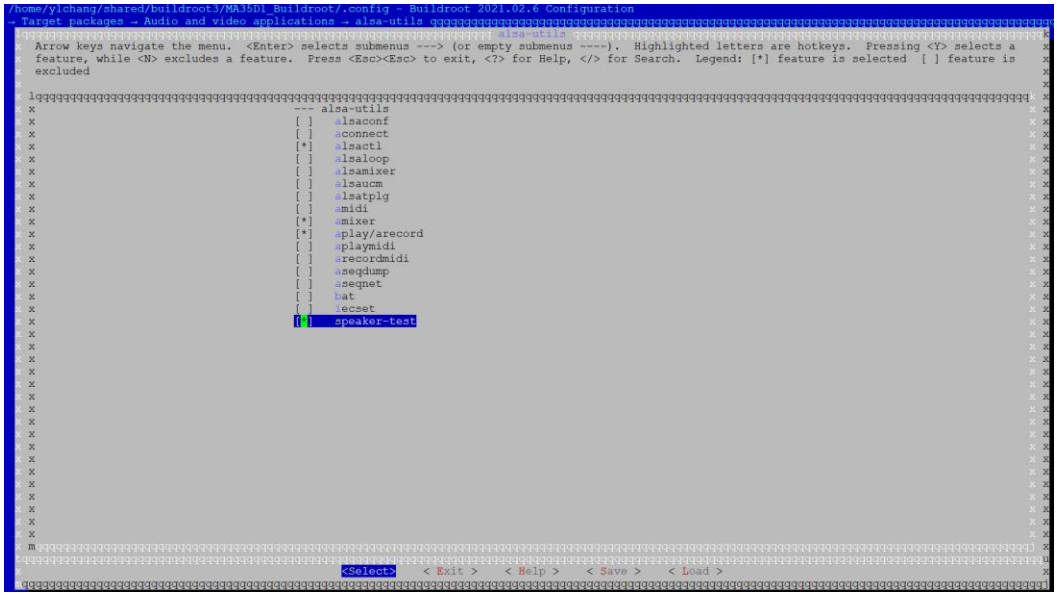
- **Simple-Card Function Test: Playback & Record**
  - Enable alsamixer & aplay
    - make menuconfig
    - Target package ---->
    - Audio and video applications --->



    - [*]alsa-utils ---->



- Press “Space” and Select “alsamixer”, “amixer”, “aplay/arecord”, and “speaker-test”
- Finally, exit and save configuration, then build the kernel image.



- Check Sound Card Registration
  - “aplay -l” or “cat /proc/around/cards”

```
# aplay -l
**** List of PLAYBACK Hardware Devices ****
card 0: nau8822hifi [nau8822-hifi], device 0: i2s_pcm-nau8822-hifi nau8822-hifi-0 [i2s_pcm-nau8822-hifi nau8822-hifi-0]
  Subdevices: 1/1
  Subdevice #0: subdevice #0
#
#
#
# cat /proc/asound/card
card0/   cards
# cat /proc/asound/card
card0/   cards
# cat /proc/asound/cards
 0 [nau8822hifi    ]: nau8822-hifi - nau8822-hifi
                      nau8822-hifi
```

- Playback Function Test
  - aplay <audio file>

```
#
# aplay hotel_california.wav
Playing WAVE 'hotel_california.wav' : Signed 16 bit Little Endian, Rate 44100 Hz, Stereo
```

- Record Function Test
  - arecord -c 2 -f S16_LE -r 48000 test.wav (record a segment of audio)
  - aplay test.wav (played the test.wav and confirmed that the sound was normal)

```
# arecord -c 2 -f S16_LE -r 48000 test.wav
Recording WAVE 'test.wav' : Signed 16 bit Little Endian, Rate 48000 Hz, Stereo
^CAborted by signal Interrupt...
# aplay test.wav
Playing WAVE 'test.wav' : Signed 16 bit Little Endian, Rate 48000 Hz, Stereo
```

- Speaker Test
  - speaker-test -c 2 -t wav -l 1 (check if sound output is normal)

